

Birmingham Active Travel Network

A GIS-based mapping and network continuity review of cycling and walking infrastructure

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1. Purpose

This study maps the existing cycling and walking (“active travel”) infrastructure across central Birmingham using open-source geospatial data, and applies a simple network-continuity screening method to flag candidate discontinuities for further investigation. It is intended as a worked demonstration of GIS data acquisition, spatial analysis, and interactive cartographic communication, produced as part of an independent portfolio project.

2. Data and Method

2.1 Data source

Infrastructure geometry and attributes were extracted from OpenStreetMap (OSM) via the Overpass API, covering a bounding box over central Birmingham (52.4650–52.4950°N, -1.9450–1.8850°W). Features were selected using the following tag criteria:

- highway=cycleway — segregated cycle paths
- cycleway=* (excluding 'no') — on-road cycle lanes
- highway=path with a bicycle tag — shared cycle/foot paths
- highway=footway with foot=designated — designated footways

OSM is a volunteer-maintained dataset. Coverage reflects mapping activity to date and should not be read as a complete inventory of physical infrastructure; some features may be missing, and attribute completeness (surface, lighting, segregation) varies by segment.

2.2 Network continuity screening

For each mapped segment, the start and end coordinates were compared against the endpoints of all other segments using the haversine great-circle distance. Any endpoint whose nearest neighbouring endpoint (on a different way) exceeded 40 metres was flagged as a candidate discontinuity. Endpoints within approximately 150m of the study area boundary were excluded to avoid false positives created by the bounding-box cutoff itself.

This is a screening heuristic, not a validated finding. A flagged point may indicate: a genuine gap in dedicated provision; a legitimate transition onto a general traffic street not captured in this extract (since ordinary roads were not queried); or a mapping omission in OSM. Field verification or cross-referencing against Birmingham City Council's own cycling infrastructure records (e.g. the Birmingham Cycle Revolution programme) would be needed to distinguish these before any of the flagged points inform investment decisions.

3. Key Findings

Metric	Value
Study area	Central Birmingham (approx. 5km × 3.3km)
Total mapped active travel network	46.0 km
Segregated cycle paths	39.5 km (85.9%)
Shared cycle/foot paths	3.1 km (6.8%)
On-road cycle lanes	1.4 km (3.1%)
Designated footways	2.0 km (4.3%)
Mapped way segments	538
Candidate network discontinuities (>40m)	91

- Segregated cycle paths dominate the mapped network (85.9% of total length), concentrated around central canal towpaths and the inner ring road.
- On-road cycle lanes make up a small minority (3.1%) of mapped provision, consistent with Birmingham's historical emphasis on off-carriageway canal-based routes over on-street lanes.
- 91 of 1,076 segment endpoints (8.5%) sit further than 40m from any other mapped segment end, after excluding boundary artefacts — a starting shortlist for targeted field review.

4. Limitations

- OSM coverage is volunteer-driven and uneven; absence of a feature in this dataset does not confirm its absence on the ground.
- The continuity screen only considers active-travel-tagged features; it cannot distinguish a genuine gap from a safe transition onto an ordinary street.
- The 40m threshold is a simplifying assumption, not a design standard; DfT's LTN 1/20 does not specify a maximum acceptable discontinuity distance in this form.
- Attribute data (surface, lighting) is incomplete for a meaningful share of segments and was not used to weight the analysis.

5. Recommended Next Steps

- Cross-reference the 91 candidate discontinuities against Birmingham City Council's cycling infrastructure GIS layers to filter out false positives.
- Extend the study area to the wider ring road and radial corridors to assess orbital and radial network coherence, not just the core.
- Layer in Index of Multiple Deprivation (IMD) and school/employment site data to prioritise investment where network gaps coincide with need.
- Validate a sample of flagged points in the field to establish a false-positive rate for the screening method.

6. Data and Licensing

Base data © OpenStreetMap contributors, available under the Open Database Licence (ODbL). Extracted via the Overpass API (overpass-api.de) on 29 July 2026. All analysis and the accompanying interactive map are original work produced for this study.